

Pelin Zeynep Kaya

Department: Rank 1st

Bachelor of Science in Computer Engineering

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PROFESSIONAL SUMMARY

Fourth-year Computer Engineering student ranked 1st in the department with a 3.88/4.00 CGPA, focused on backend software engineering and applied AI. Built secure full-stack systems and machine learning projects using Python, C#, Java, Node.js, Spring Boot, ASP.NET Core, and PostgreSQL. Currently conducting biomedical AI research on melanoma drug-response prediction and studying parameter-efficient fine-tuning methods including LoRA and QLoRA. Seeking a long-term AI or software engineering internship where I can contribute to real products, learn from experienced engineers, and continue after graduation.

EDUCATION

•Bachelor of Science in Computer Engineering

Oct 2022 - Jun 2027

Istanbul Arel University, Istanbul, Turkey

CGPA: 3.88/4.00 | Department Rank: 1st

EXPERIENCE

•Automation Intern

Aug 2026

Akçansa – Betonsa Technology Center, Technical Affairs Directorate

- Supported software-oriented work in the Automation Unit, examining PLC/HMI communication, industrial network configuration, and troubleshooting workflows.
- Developed the initial architecture for a Python/OpenCV computer-vision prototype, structuring camera capture and detection components for later integration with automation systems.

ACHIEVEMENTS & HONORS

•Department Rank: 1st & High Honor Student

2022 - Present

Computer Engineering Department, Istanbul Arel University

- Consistently awarded High Honor Certificate every semester, maintaining a 3.88/4.00 CGPA and ranking 1st in the department.

•100% Merit Scholarship

2022 - Present

Awarded for continuous outstanding academic excellence

- Secured full tuition coverage based on high-tier academic performance and institutional tracking.

•Datathon 2026 Team Participant

2026

Data Science, Algorithmic Problem Solving & Optimization Track

- Participated as part of a team, collaborating on data analysis and algorithmic problem-solving tasks under time constraints.

RESEARCH & APPLIED AI

•EEG Stationarity Analysis & Generative XAI 🐙 GitHub

2025 - 2026

PyTorch, Diffusion Transformers, Signal Processing, Explainable AI

- Conducted a comparative stationarity analysis between 19-channel pediatric ADHD EEG signals and healthy adult control datasets, adapting the research scope to overcome the unavailability of matching adult ADHD data.
- Engineered a custom Diffusion Transformer architecture, implementing specialized training schedulers and embeddings to model and synthesize complex clinical time-series data.
- Utilized Explainable AI (XAI) methodologies to produce frequency-domain proofs, validating the physiological integrity and temporal dynamics of the generated signals.

•Melanoma Drug-Response Prediction Research

2026 - Present

Python, GDSC2, Gene Expression Analysis, Scikit-learn, XAI

- Collaborating with a Molecular Biology and Genetics student to develop a literature-supported gene panel and machine-learning pipeline for predicting melanoma response to BRAF and MEK inhibitors.
- Analyzing gene-expression and fitted dose-response data, with planned feature-selection and explainability experiments to connect predictions with biologically meaningful pathways; targeting a conference submission.

•Parameter-Efficient LLM Fine-Tuning Study

2026 - Present

Transformers, LoRA, QLoRA, PEFT, PyTorch

- Preparing an implementation-oriented Medium article based on research papers and technical documentation that compares LoRA and QLoRA across quantization, memory trade-offs, training constraints, method-selection criteria, practical limitations, and reproducible fine-tuning workflows.

SELECTED PROJECTS

•Aura Board: Deployed AI-Assisted Productivity Platform [GitHub](#) [Live Demo](#) 2026

C#, ASP.NET Core, PostgreSQL, JavaScript, Gemini API

- Developed and deployed Aura Board, a full-stack ASP.NET Core and PostgreSQL web application for creating and managing personalized vision boards.
- Implemented authentication and session management using JWTs stored in HttpOnly cookies, integrated the Gemini API through backend services, and added PDF export functionality for board sharing.

•Hybrid SSH Tunneling: Secure Communication Protocol [GitHub](#) 2026

C++, Cryptography, Diffie-Hellman, Socket Programming

- Developed a client-server communication system in C++ using socket programming to simulate secure packet transmission.
- Implemented Diffie-Hellman key exchange and custom encryption workflows to establish secure communication channels between network nodes.

•Overengineering Detector: Architecture Analysis Platform [GitHub](#) 2026

Node.js, Express.js, PostgreSQL, Vanilla JS, OpenAPI, Swagger

- Designed a Node.js backend application that evaluates software architecture decisions through a deterministic rule-based scoring engine.
- Implemented REST APIs, JWT authentication, role-based access control (RBAC), and PostgreSQL persistence for user and project management.

•CodeYourTree: Gamified Habit-Tracking Platform [GitHub](#) 2026

Java 21, Spring Boot, Spring Security, PostgreSQL, HTML5 Canvas

- Developed a habit-tracking application using Spring Boot, Spring Security, and PostgreSQL with JWT-based authentication.
- Designed REST APIs for user and progress management and implemented visual streak tracking using HTML5 Canvas.

•StegoDetector: Hybrid Steganalysis & Threat Detection [GitHub](#) 2026

Python, Streamlit, XGBoost, Scikit-learn, OpenCV, NumPy

- Developed a hybrid steganalysis system to distinguish hidden-information images from normal images using machine learning.
- Engineered a Soft Voting Ensemble combining XGBoost and Random Forest models, achieving 80% classification accuracy.
- Built feature extraction and visualization pipelines with OpenCV, NumPy, and Streamlit for model evaluation and diagnostics.

TECHNICAL SKILLS

Programming Languages: Python, C#, Java, SQL, C++, JavaScript, TypeScript

Backend & Databases: ASP.NET Core, Node.js, Express.js, Spring Boot, FastAPI, REST APIs, PostgreSQL, Database Design, OpenAPI, Swagger, JWT

AI & Machine Learning: PyTorch, Scikit-learn, XGBoost, Transformers, LoRA, QLoRA, PEFT, Explainable AI (XAI), Feature Engineering, Model Evaluation

Data & Computer Vision: Pandas, NumPy, OpenCV, Data Visualization, Signal Processing, Gene Expression Analysis

Software Engineering: Git, Agile, Scrum, Software Development Lifecycle (SDLC), Requirements Analysis, System Design, UML

LEADERSHIP & PROJECT LEADERSHIP

•CampusNote Pro [GitHub](#) 2026

Product Owner & Scrum Master (Academic Team Project)

- Coordinated sprint planning, backlog prioritization, and task assignment for a multidisciplinary student software team.
- Facilitated communication between development activities and project requirements to support timely feature delivery.
- Contributed to the implementation of AI-assisted platform features using Python to support the core academic sharing ecosystem.